

Fan-Assistant.md

Fan Assistant

ArenaMind AI UI/UX Specification

Screen ID: FAN-001

Version: 1.0

Status: Draft

Priority: ★★★★★

Overview

The **ArenaMind AI Fan Assistant** is a personalized AI companion designed to guide spectators throughout their entire FIFA World Cup match-day journey.

It combines:

-  Generative AI
-  Indoor Navigation
-  Multilingual Assistance
-  Real-time Location Intelligence
-  Personalized Recommendations
-  Voice Interaction
-  Vision AI
-  Accessibility Support

Unlike traditional event apps, the Fan Assistant understands natural language and proactively helps fans before they encounter problems.

Product Vision

"Every fan should feel like they have a personal AI concierge inside the stadium."

The Fan Assistant delivers contextual, real-time guidance from home to stadium and back again.

User Personas

Primary Users

- Football Fans
 - International Visitors
 - Families
 - First-time Stadium Visitors
 - VIP Guests
 - Accessibility Users
-

Primary Goals

The Fan Assistant should help users:

- Find their seat
 - Navigate the stadium
 - Avoid crowded areas
 - Discover food and merchandise
 - Receive transport guidance
 - Access emergency assistance
 - Communicate in their preferred language
-

User Journey

```
Home
|
Travel Planning
|
Arrival
|
Entry Gate
|
Seat Navigation
|
Food & Merchandise
|
Live Match
|
Exit Planning
|
Transport Home
```

ArenaMind AI assists throughout the entire journey.

Layout Structure

Match Header + Ticket + Language
AI Greeting
Smart Search / Voice / Camera
Personalized Quick Actions
Interactive Stadium Map
AI Recommendations
Live Notifications

Welcome Section

Displays

Welcome, Vaibhav 🖐️

FIFA World Cup 2026

Argentina 🇦🇷

vs

Brazil 🇧🇷

Kickoff

19:30

Seat

B12 • Row 18

Gate

A

The assistant immediately provides contextual suggestions.

AI Greeting

Generated dynamically.

Example

Welcome!

Your match begins in 1 hour.

Gate A currently has moderate traffic.

Walking through Gate C will save approximately 8 minutes.

Rain is expected later this evening.

Would you like a personalized route?

Smart Input

Supports

- Text
- Voice
- Camera
- Image Upload

Placeholder

Ask ArenaMind...

"Where is my seat?"

"Find vegetarian food."

"Take me to the nearest restroom."

"Translate this sign."

Quick Actions

Adaptive shortcuts.

Examples

 My Ticket

 Navigate

 Food

 Merchandise

 Restrooms

 Transport

& Accessibility

 Emergency

The order changes based on context.

Interactive Stadium Map

Primary navigation experience.

Supports

- Indoor Navigation
 - Live Position
 - Turn-by-turn Guidance
 - AR-ready architecture (future)
 - Dynamic Route Updates
-

Map Layers

Seat

Gate

Food

Medical

Restroom

Parking

Transport

Accessibility

Emergency

Merchandise

Users can toggle layers independently.

Smart Navigation

Example

User

Take me to my seat.

ArenaMind AI

Walking Time

4 min

Distance

210 m

Current Route

Gate A

↓

Section B

↓

Row 18

↓

Seat 12

If congestion occurs, the route updates automatically.

Food Discovery

The assistant recommends nearby food options.

Cards include

- Name
- Cuisine
- Queue Time
- Distance
- Price Range
- Dietary Tags

Example

Stadium Grill

★ 4.7

2 min away

Queue

6 min

Vegetarian Available

Merchandise Explorer

Displays

- Official Stores
- Exclusive Match Items
- Queue Estimates
- Distance

AI recommends less crowded stores.

Live Crowd Guidance

The assistant proactively warns users.

Example

Crowd Alert

Gate A is becoming crowded.

Recommendation

Use Gate C instead.

Estimated Time Saved

7 minutes.

AI Recommendations

Context-aware suggestions.

Examples

Kickoff begins in 15 minutes.

Proceed to your seat now.

Halftime approaching.

Food Court B has the shortest queue.

Heavy rain expected.

Covered walkway recommended.

AI Vision

Users can use their camera to identify stadium features.

Examples

What is this entrance?

Read this sign.

Find my seat.

Translate this board.

The assistant overlays guidance on the image.

Multilingual Support

Supports

- English
- Spanish
- French
- Portuguese
- Arabic
- Hindi
- Japanese
- Korean
- German
- Italian
- Chinese

Real-time translation for

- Voice
- Text
- Announcements

- Signage
-

Accessibility Copilot

Dedicated mode for accessible navigation.

Supports

- Wheelchair-friendly routes
 - Elevator guidance
 - Hearing assistance
 - Voice-first navigation
 - Large text
 - High contrast
-

Emergency Assistance

Persistent floating button.

Capabilities

Call Medical

Locate Security

Share Live Location

Emergency Exit

Contact Volunteers

AI automatically provides the safest evacuation route when needed.

Transport Assistant

After the match, ArenaMind AI recommends:

- Metro
- Bus
- Taxi
- Ride-share
- Walking
- Parking Exit

Example

```
Best Option
Metro Line 2
Departure
12 min
Walking
6 min
Expected Crowd
Low
```

AI Conversation Panel

Supports continuous conversations.

Example

```
User
Where can I buy a jersey?
↓
ArenaMind
Official Store A
```

5 minutes away

Queue

8 minutes

Would you like directions?

Conversation context is preserved.

Notification Center

Examples

- Kickoff Reminder
- Gate Change
- Weather Alert
- Transport Delay
- Crowd Alert
- Emergency Broadcast

Notifications are contextual and actionable.

Personal Match Timeline

Travel

↓

Arrival

↓

Entry

↓

Seat

↓
Kickoff
↓
Halftime
↓
Full Time
↓
Exit
↓
Transport

Completed milestones are highlighted.

Component Inventory

- Match Header
- Ticket Card
- AI Greeting
- Smart Search
- Voice Button
- Camera Button
- Quick Action Grid
- Interactive Stadium Map
- Recommendation Cards
- Crowd Alerts
- Notification Center

- AI Chat Panel
 - Emergency FAB
 - Transport Widget
-

Motion Design

Greeting

Fade + Slide

Map

Smooth pan + zoom

Recommendations

Slide up

Notifications

Fade + glow

Route Changes

Animated path transition

Voice

Waveform animation

AI Responses

Streaming text + confidence reveal

Accessibility

Supports

- Voice Navigation
- Screen Readers
- Keyboard Navigation
- High Contrast

- Reduced Motion
 - Large Text
 - WCAG 2.2 AA
-

Responsive Design

Mobile (Primary)

Single-column layout.

Bottom navigation.

Persistent AI input.

Floating emergency button.

Tablet

Split view.

Map + recommendations.

Desktop

Expanded map with AI panel.

Empty States

If location is unavailable

Enable location to receive
personalized navigation.

If no recommendations

Enjoy the match!

No immediate actions required.

Loading Experience

Display

- Stadium outline
- AI pulse
- Skeleton cards
- Route animation

No generic spinners.

Error Handling

If Gemini exceeds rate limits

AI service switching...

Powered by Groq.

Conversation preserved.

The transition must be invisible to the user.

Performance Targets

Metric	Target
Initial Load	<2 s
AI Response	<3 s
Map Interaction	<100 ms

Metric	Target
Route Update	<500 ms
Lighthouse Score	90+

Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS
- Framer Motion

Maps

- Google Maps Platform
- Indoor Maps

AI

- Gemini 2.5 (Primary)
- Groq (Fallback)
- Vertex AI
- Vision AI
- Speech-to-Text
- Text-to-Speech
- Translation API
- RAG Engine

Backend

- FastAPI
- Firebase

- Firestore
 - WebSockets
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Acceptance Criteria

- Personalized greeting appears on launch.
 - Indoor navigation updates dynamically.
 - AI responds through text and voice.
 - Camera identifies stadium objects and signage.
 - Multilingual translation works across supported languages.
 - Crowd alerts update in real time.
 - Transport recommendations adapt after the match.
 - Gemini-to-Groq failover is seamless.
 - Mobile experience remains intuitive and responsive.
 - Accessibility requirements satisfy WCAG 2.2 AA.
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Future Enhancements

- AR indoor navigation
 - Apple Vision Pro spectator mode
 - Smart wearable integration
 - Personalized souvenir recommendations
 - Live friend tracking
 - AI-powered match highlights
 - Stadium gamification
 - Loyalty and rewards
 - Digital collectibles
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Fan Assistant Philosophy

"Fans shouldn't have to search for information—the stadium should understand what they need before they ask."

ArenaMind AI transforms the spectator journey into a personalized, multilingual, AI-guided experience that removes friction and allows fans to focus on what matters most: enjoying the world's biggest football tournament.

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